Higher National Diploma in Civil Engineering 3

2nd Year, Second Semester Examination - 2022

CE2215-Structural Analysis

Questions

1. Virtual Work/Energy Method

I. Write the **principle of virtual work** for virtual displacement (△) and rotational displacement () in terms of standard notations. Also, annotate the notations. (08 Marks)

II. A simply supported beam AB of span **2L**, carries a **UDL of 3wkN/m$** as shown in Figure 01. Determine the **vertical deflection at midpoint C** and **rotational displacement (slope) at point A** of the beam using the **virtual work method**. (12 Marks)

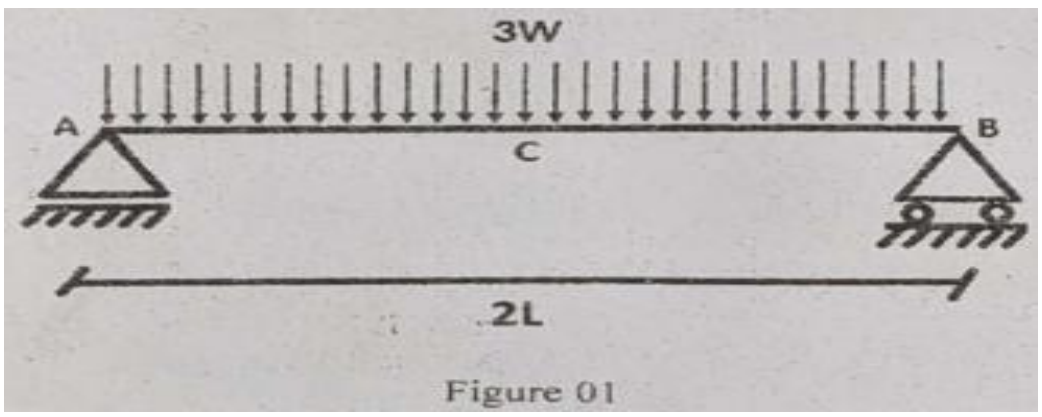

Figure 01

2. Three Moment Theorem

I. State the **three-moment equation** of a two-span continuous beam with the aid of a sketch and annotate the notations used in the equation. (06 Marks)

II. A continuous beam ABCD is subjected to UDLs and a point load as shown in Figure 02. Compute the **bending moments M_A, M_B, M_C, and M_D** using a **three-moment theorem** at supports A, B, C, and D respectively. Also, plot the bending moment diagram. Assume EI is constant throughout the beam. (14 Marks)

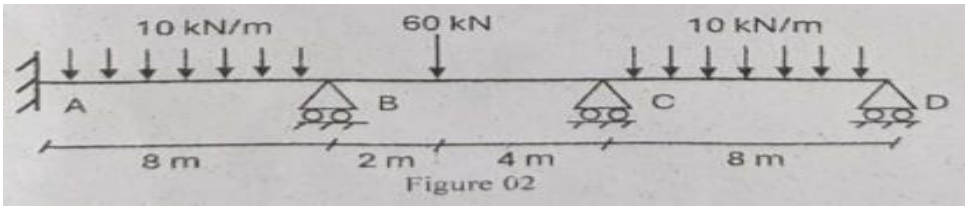

Figure 02

3. Slope Deflection Method

I. State all the **assumptions** which are used in the **slope deflection method**. (04 Marks)

II. Compute the **bending moment** of the continuous beam ABC as shown in Figure 03 using the **slope deflection method** and plot the **bending moment diagram**. Assume that EI is constant throughout the beam. (16 Marks)

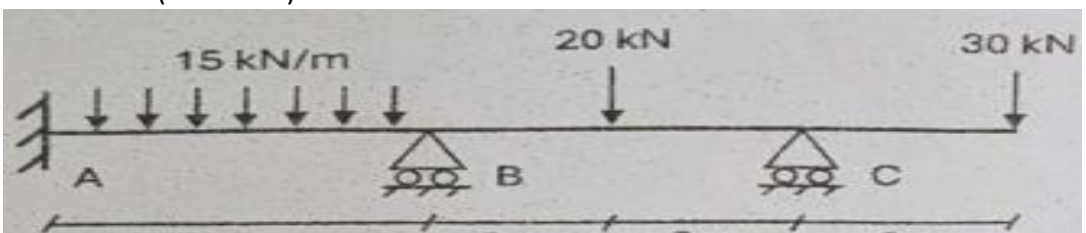

Figure 03

4.

I. Briefly explain why truss members carry **only axial forces** but not bending moments. (02 Marks)

II. Discuss the **two cases** where member forces are **zero (zero force member)** in trusses with the aid of a sketch. (06 Marks)

III. Analyze the truss as shown in Figure 04 by using the **method of section** and tabulate the **member forces of BC, BF, and EF** with their action (Tension or Compression). (12 Marks)

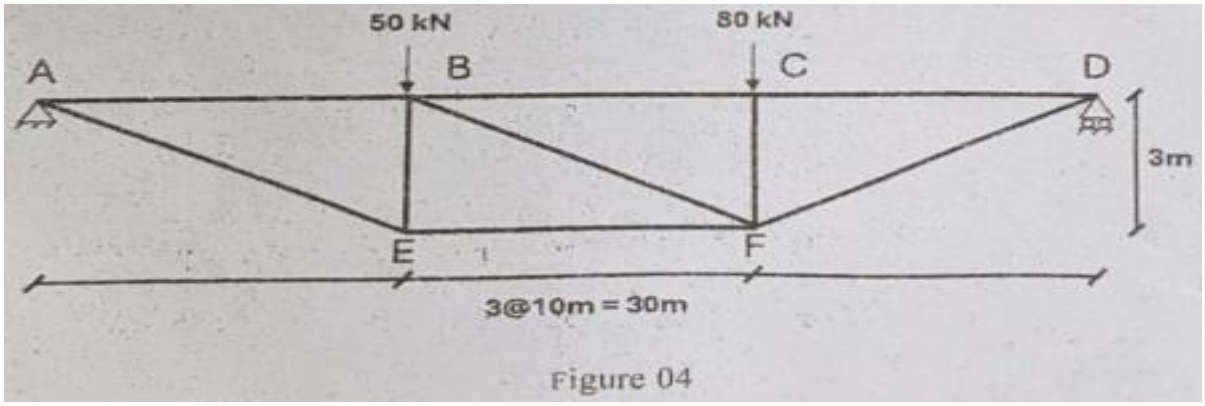

Figure 04

5. Moment Distribution Method

A Continuous beam ABCD carries UDLs of 4kN/m and a point load of 15kN as shown in Figure 05.

I. Determine the **moment distribution factors**. (06 Marks)

II. Analyze the beam ABCD using the **moment distribution method** and find the **bending moment** of the beam ABCD. Also, plot the bending moment diagram. Assume that EI is constant throughout the beam. (14 Marks)

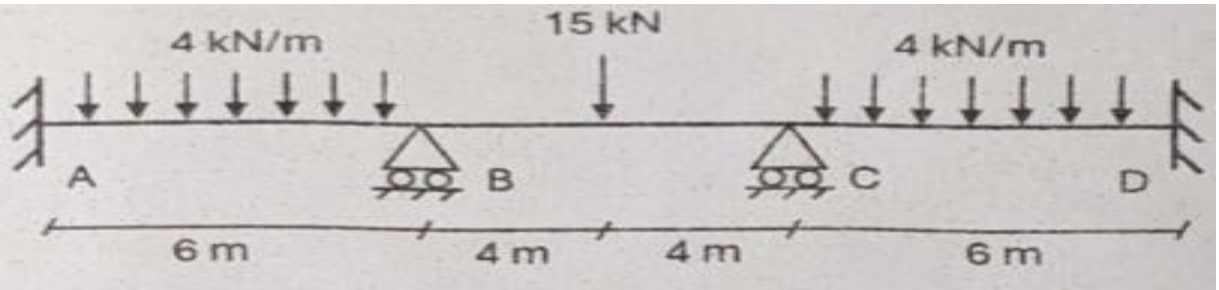

Figure 05

6.

Find the **horizontal reactions** of a **two-hinged parabolic arch** as shown in Figure 06. Neglect the shear and axial forces. (20 Marks)

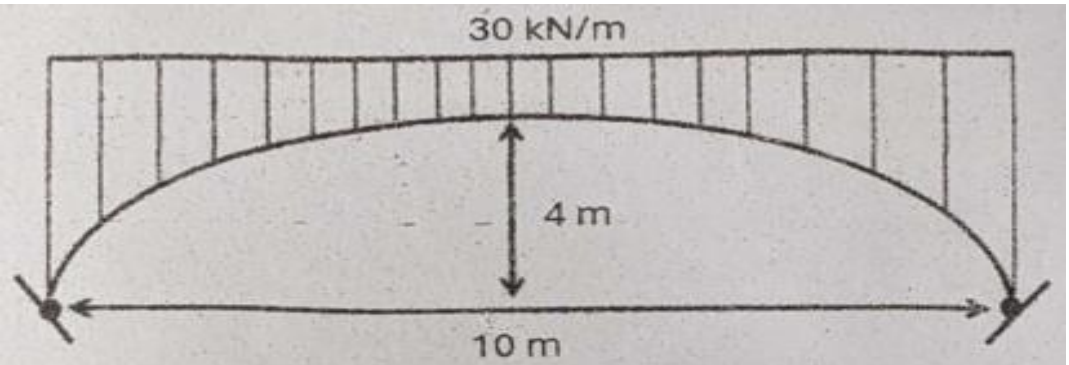

Figure 06

For parabolic arch $y_{(x)} = \frac{4hx(L-x)}{L^2}$ with usual notations